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Cybersecurity
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Ethical hacking

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Chapter 1

Internet

1.0.1 What is internet

Internet is a network of networks. There are hierarcical networks such as:

- Internet backbone: connecting the ISPs' backbones
- ISP backbone: connecting organizations' backbones
- Organization backbone connects local area networks (LANs)
- LAN connects end systems

Internet is diveded in:

- Access network: connects end systems to the first router (e.g., DSL, cable, fiber, wireless)
- Physical media: carries signals between devices (e.g., twisted pair, coaxial cable, fiber optic, radio waves)
- Network core: connects the access networks (e.g., routers and links). Routers work togheter to figure a path for each packet from source to destination. The routing table are mantained in real time and a routing algorithm can adapt to changing internet conditons.

1.0.2 protocols

Protocols specify rules about the service, such as:

- procedure rules: types and sequence of messages exchanged, syntax and semantics, action to take in respect of messages and events
- message format size and coding
- timing to wait between events, timeouts, flow control

1.0.3 Access layer

Its constituted by network with end points of the same local management, provides connectivity among station on the same network and they can comunicate directly among them.

Ethernet(IEEE 802.3) is the most common access control protocol. Each host has a NIC (Network Interface Card) with a fixxed address. MAC addresses are 48bits long and uniquely identify the host in the network. Each ethernet frame has a fixed format.

Figure 1.1: Ethernet Frame Format

All the host connected to the same transmission system are a nework. An ethernet network costitutes a brodcast domani, where, ideally, frames sent in that domain are recived by all the host. Switches limit the explosion of the packets. Only the brodcast messages are replicated. Switches remembers the sourca MAC address on different ports. They only replicate the frame on the segment where the MAC address replies. Ethernet is inefficent in large network, as it uses brodcasst packets.

1.1 IP

1.1.1 IP Addresses

There are two version of IP: IPv4(32bit) and IPv6(128bit).

There are three types of IP addresses:

- Unicast: identifies a single host on the network
- Multicast: identifies a group of hosts on the network
- Broadcast: identifies all hosts on the network

there are classes of IP addresses:

- Class A: 24 bit for host.
- Class B: 16 bit for host.
- Class C: 8 bit for host.
- Class D: Used for multicast addressing.
- Class E: Reserved for future use.

There are routable and non routable IP addresses. The routable have to be unique on the internet, whike non routable can exist in different network but can not be used on internet.

1.2 Layering

the comunication between hosts in the network is organized in task assigned to a layer. Each layer offers a service to the layer above and uses the services of the layer below. the data is trasfered from the application level to application level over a network. Each layer adds protocol information and data to the layer below. The physical layer send the data physically to the destination. In the destination, the phisical layer sends the data up the stack. Each protocol reads the appropriate protocol information ad forward the data to the layer above.

There are 2 main layering models:

- ISO/OSI model: 7 layers (Physical, Data Link, Network, Transport, Session, Presentation, Application)
- TCP/IP model: 4 layers (Link, Internet, Transport, Application)

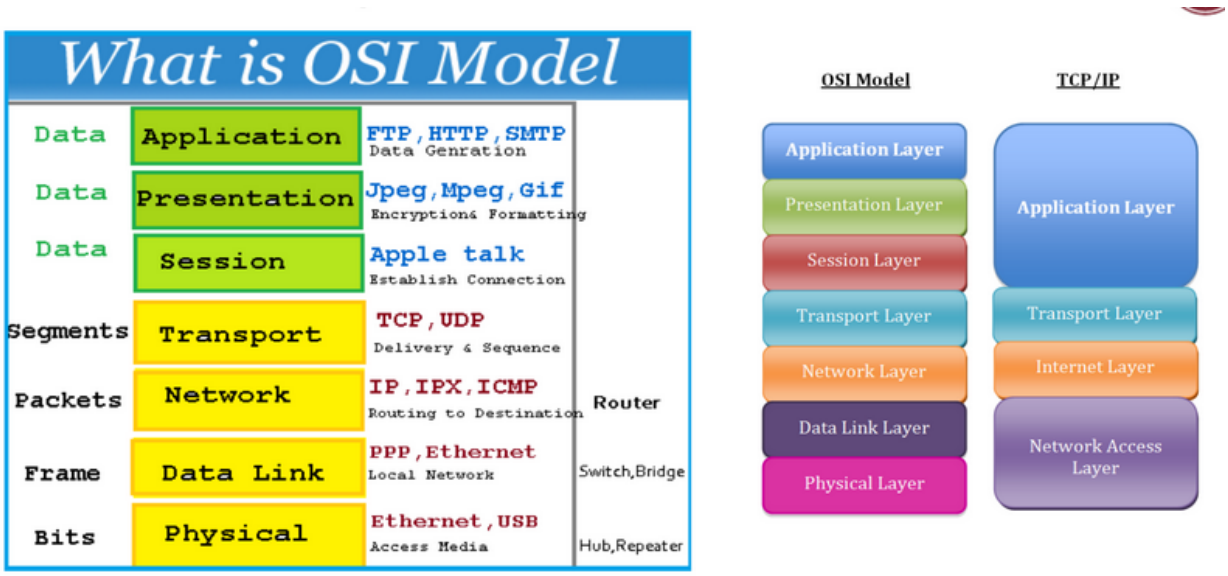

Figure 1.2: Models

1.2.1 TCP/IP model

- application layer: Corresponds to the top three layers of the OSI model.(Protocols: SMTP (sending e-mail), HTTP (web), FTP (file transfer), and others). Address: internet name (e.g., www.google.com)
- Transport layer: Equivalent to Layer 4 (Transport) of the OSI model.(Protocols: TCP, UDP). Address: port number (e.g., 80 for HTTP)
- Internet: Equivalent to layer 3 (network) of the OSI model.(Protocols: IP, ICMP, IPSec). Address: IP address (e.g., 192.168.1.1)
- Datalink: Equivalent to layer 2 (data link) of the OSI model.(Protocols: Ethernet, WiFi, ARP, etc.). Address: MAC address (e.g., 00:0a:95:9d:68:16)
- Physical layer: Equivalent to Layer 1 (Physical) of the OSI model.

Bibliography